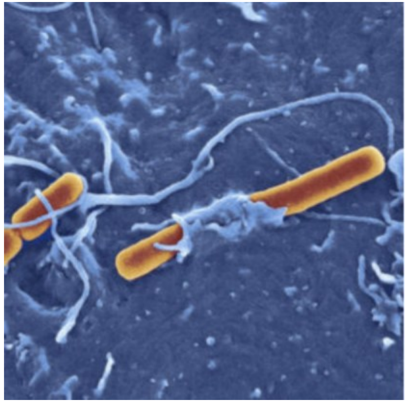


Genomic & Epidemiological Surveillance of Bacterial Pathogens

22 – 26 June 2026

Module 1: Sequencing & QC

wellcome connecting science

Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

COURSE MANUAL ON GITHUB

https://github.com/WCSCourses/Bacterial_Genomic_Surveillance_2026

wellcome connecting science

Bacterial Genomic Surveillance - Ireland and Northern Ireland

22–26 June 2026, Department of Microbiology, Moyne Institute of Preventive Medicine, Trinity College Dublin, Ireland

Wellcome Connecting Science Course Run Website
Course manual
Course Time Table 2026
Course Informatics Guide

Integrating epidemiology and genomics for surveillance and disease control

Summary

In the last few years, the introduction of advances in genomics, sequencing technologies, and our ability to interpret high resolution sequencing data for surveillance purposes – both regionally and internationally – lead to a shift from the use of molecular typing data to using Whole Genome Sequencing (WGS) for epidemiological surveillance and

wellcome connecting science

Bacterial Genomic Surveillance - Ireland and Northern Ireland

Course Manual Dublin 2026

- Genomic Capacity Development for Public Health Settings
- World Café: Translating Genomics into Public Health Action
- Module 1: Sequencing & QC
- Module 2: Genome Visualisation Tools
- Module 3: Comparative Genomics (lecture only)
- Module 4: Mapping and Phylogenetics
- Module 5: Genome Assembly and Annotation
  - Part 1: Assembly Methods Comparison
  - Part 2: Genome Annotation
- Introduction to epidemiology and genomic collaborative surveillance
- Module 6: Web tools for genomic epidemiology
- Group Task
- Appendices

COURSE MANUAL ON GITHUB

https://github.com/WCSCourses/Bacterial_Genomic_Surveillance_2026

Module 1 - Sequencing & QC - Dublin 2026

Module lead: Tomás Caride Poklepovich

Table of contents

VIRTUAL MACHINE

- A virtual machine (VM) is a "completely isolated guest operating system installation within a normal host operating system".
- For the course, you will be using a Linux (guest) operating system within a Windows or other (host) environment. So that you can switch between them.

VIRTUAL MACHINE

ADVANTAGES

- Encapsulates the entire course

- software
- course data

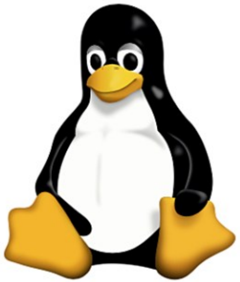
- Easily moved or copied

  - You can continue with the course and use the software back at home

- A useful feature of the VM is the ability to transfer data between the VM and host operating system using a shared folders and bidirectional file transfer.

LINUX

- Output of lots of biological research exists in the form of large text files
- Several powerful and flexible commands for processing large text files
- Freely available and flexible
- Widely used in scientific community

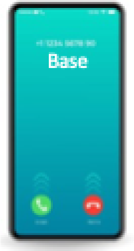


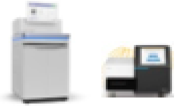
Conda environments

- Bioinformatics softwares usually rely on many dependencies (other softwares/tools)
- Different softwares/tools you need for your pipeline analysis may use different dependencies, and many times uses different versions of tools
- For example: *Tool A* uses Python 3.6 and *Tool B* uses Python 3.9
- You can create different environments for different tools or group of tools
- Widely used in scientific community, it makes easier to install software

CONDA

WHAT DO WE GET AFTER A SEQUENCING RUN





Illumina

Fastq

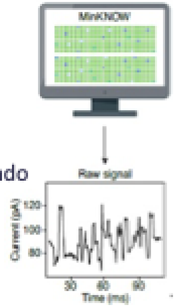
BASE CALLING[1]
y
DEMULTIPLEXING

Nanopore

Fast5/Pod5

[1]MinKNOW
Guppy/Dorado

Fastq

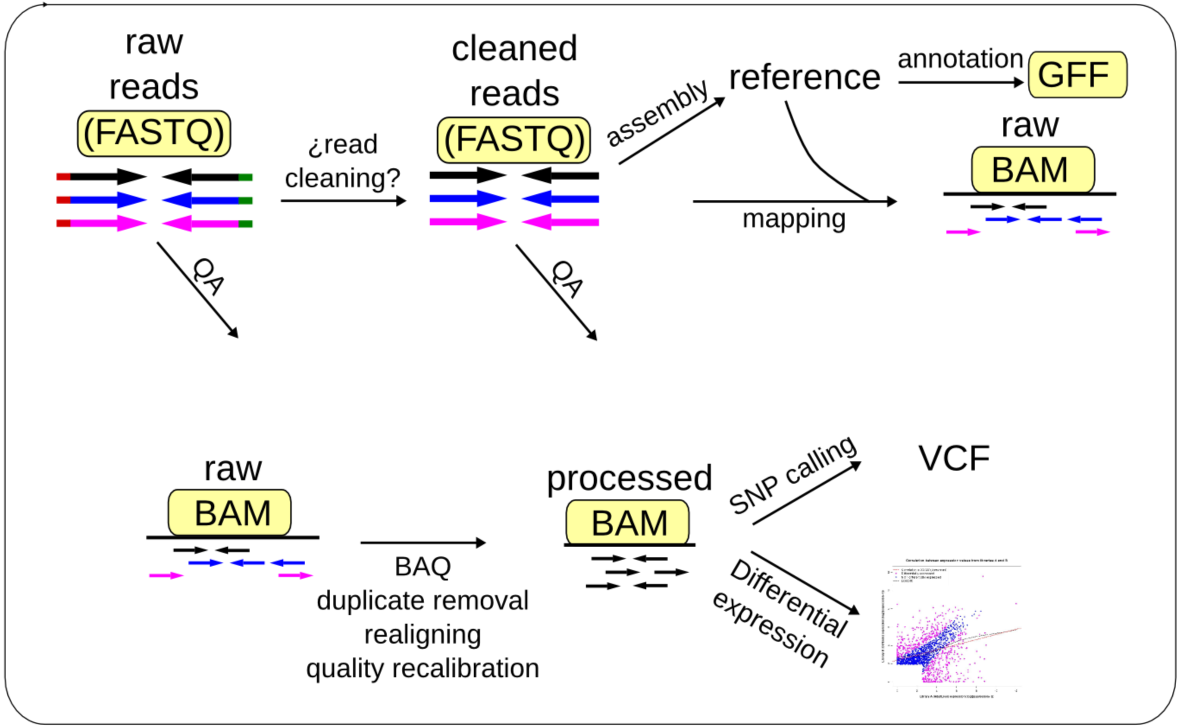


Quality control

Different bioinformatic analyses

DEMULTIPLEXING

Sequence analysis

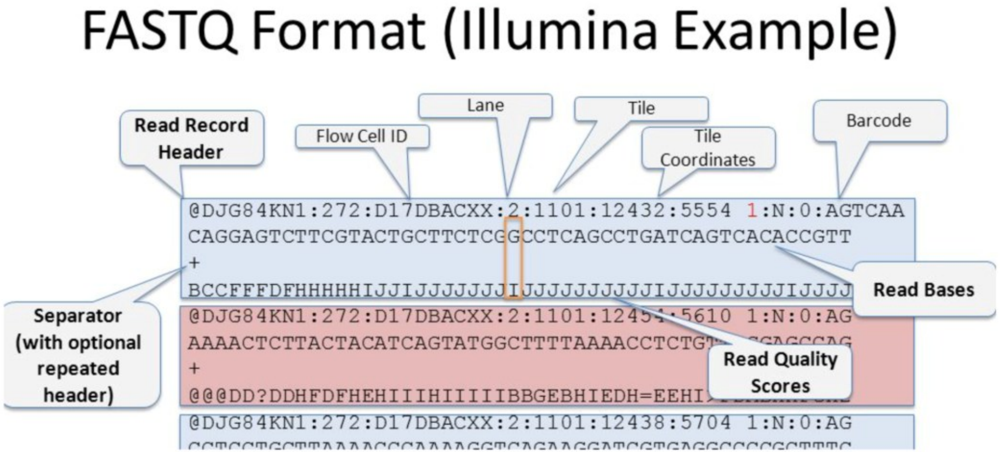


https://bioinf.comav.upv.es/courses/sequence_analysis/_images/ngs_map_read_file_formats.png

WHY DIFFERENT FILE FORMATS?

We need ways to represent our (biological) data

- Sequencing instruments:
    - FastQ
- Nucleotides *vs.* Amino acids
    - FASTA
- Alignments
    - SAM/BAM
- SNPs/InDels
    - VCF
- Genes, coding sequences and features
    - GBK/GBFF
    - GFF3
    - EMBL



MODULE 1 - OVERVIEW

1. Perform quality control (QC) assessments on FASTQ-formatted sequence data obtained from both short and long reads sequencing technologies.
2. Identify possible contamination in high throughput sequence data.
3. Critically evaluate and interpret QC outputs to determine data reliability.

- Introduction

- Practical Exercises

- Short read quality control – FastQC, Kraken2 & TrimGalore

- Long read quality control – NanoPlot, Kraken2 & Filtlong

- Bonus – MultiQC & Pavian

- Wrap up questions

FastQC

In a graphic way, it allows you to check on some basic quality parameters from reads

Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Statistics

Measure	Value
Filename	CNGB1553_S31_L001_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	528785
Total Bases	132.7 Mbp
Sequences flagged as poor quality	0
Sequence length	251
%GC	49

Per base sequence quality

$$q = -10 \times log_{10}(p)$$

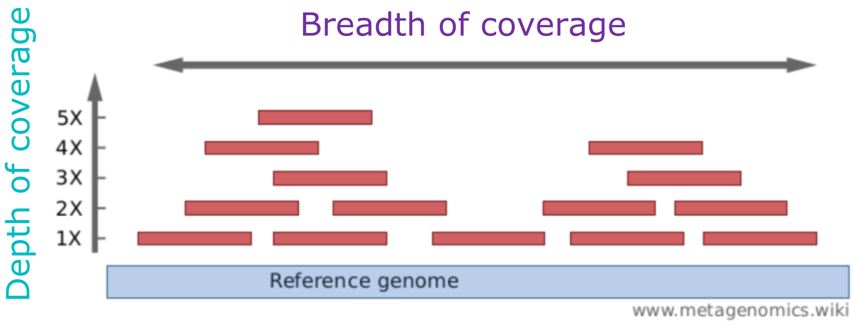
Donde:

- q = quality value
- p = estimated probability error for a base call

Ejemplos:

- q = 20 significa p = 10^{-2} (1 error cada 100 bases)
- q = 30 significa p = 10^{-3} (1 error cada 1000 bases)
- q = 40 significa p = 10^{-4} (1 error cada 10000 bases)

Depth of coverage vs breadth of coverage



Depth of coverage (mapping depth)

How many times a genome is "covered" by sequenced reads?

Mean depth of coverage is calculated as the number of bases of all short reads that match a genome divided by the length of this genome. It is often expressed as 2X, 30X, 100X,...

Breadth of coverage (covered length)

How many bases of a genome are "covered" by sequenced reads? Are there regions that are not covered, even not by a single read?

It is the percentage of bases of a reference genome that are covered with a certain depth.

For example: "90% of a genome is covered at 1X depth; and still 60% is covered at 3X depth."

Mean depth of coverage

Read 1

Basic Statistics

Measure	Value
Filename	MB39120_S21_L001_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1718971
Total Bases	224.3 Mbp
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	53

Read 2

Basic Statistics

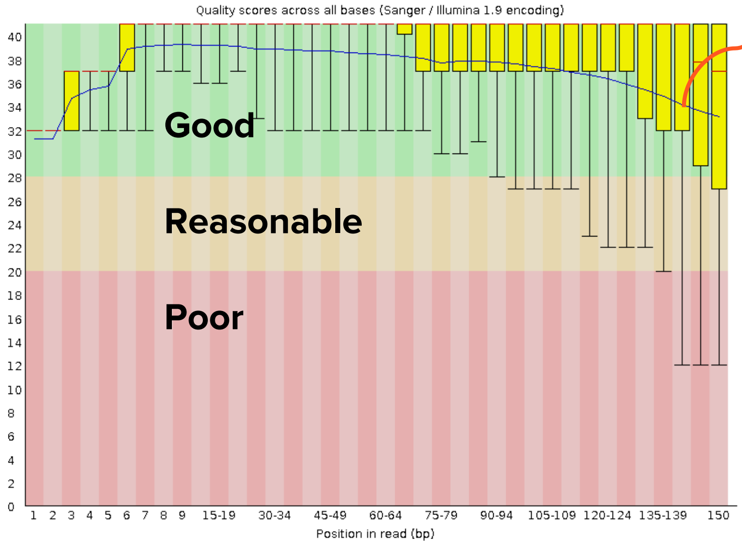
Measure	Value
Filename	MB39120_S21_L001_R2_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1718971
Total Bases	224.5 Mbp
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	53

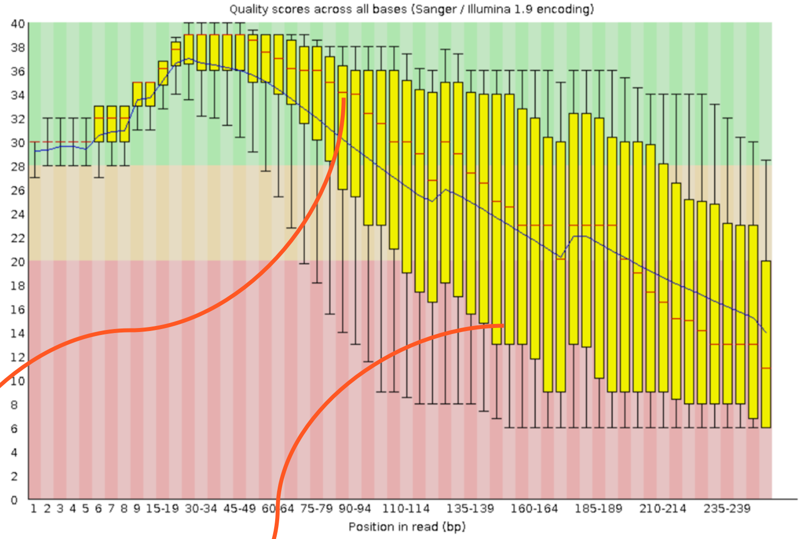
$$\text{Average depth of coverage} = \frac{\text{Total bases}}{\text{Genome size}} = \frac{224.3 + 224.5\ \text{Mbp}}{4.4\ \text{Mbp}} = 102\text{X}$$

FastQC

Per base sequence quality

mean score





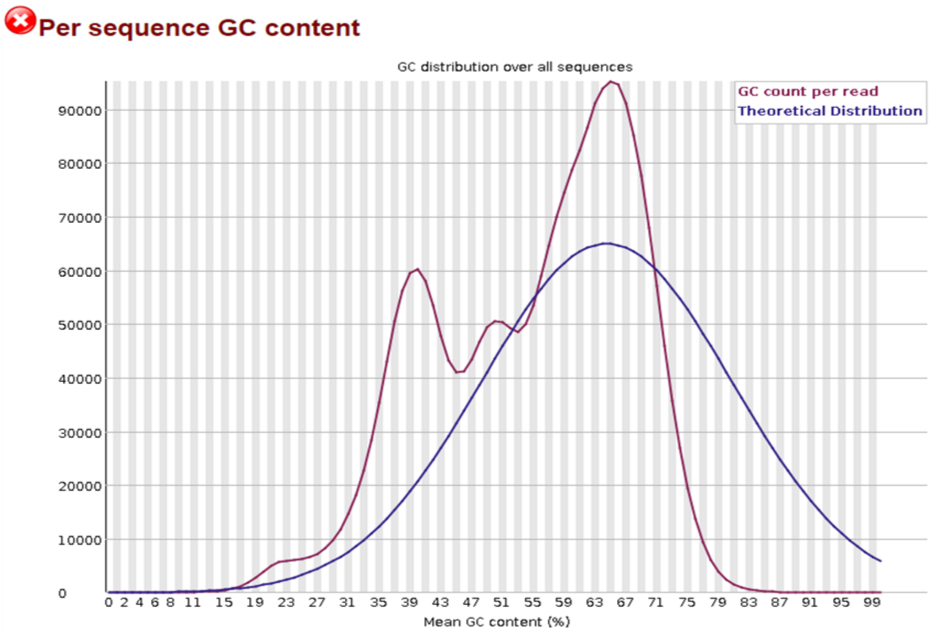
median score

inner quartile score range

- The central red line is the median value
- The yellow box represents the inter-quartile range (25-75%)
- The upper and lower whiskers represent the 10% and 90% points
- The blue line represents the mean quality

FastQC

Per sequence GC content



Kraken2

- Assigns taxonomic labels to DNA sequences using exact k-mer matches to a database of known genomes
- Useful to identify different species of reads obtained by metagenomics
- For WGS is very useful for detecting contaminations
- Careful with misidentifications!

1= % of reads covered by the clade rooted at this taxon

2= Number of reads covered by the clade rooted at this taxon

3= Number of reads assigned directly to this taxon

4= Taxonomic rank: (U)nclassified, (R)oot, (D)omain, (K)ingdom, (P)hylum, (C)lass, (O)rder, (F)amily, (G)enus, or (S)pecies

5= NCBI taxonomy ID

6= Scientific name

1	2	3	4	5	6
2.73	4508	4508	U	0	unclassified
97.27	160631	4999	-	1	root
94.24	155631	0	-	131567	cellular organisms
94.24	155630	435	D	2	Bacteria
93.91	155085	20	-	1783272	Terrabacteria group
93.90	155065	70	P	1239	Firmicutes
93.86	154993	95	C	91061	Bacilli
93.79	154889	319	O	1385	Bacillales
93.60	154566	328	F	90964	Staphylococcaceae
93.40	154237	8437	G	1279	Staphylococcus
88.28	145787	106808	S	1280	Staphylococcus aureus
23.60	38971	38723	-	46170	Staphylococcus aureus subsp. aureus
0.13	215	215	-	1074252	Staphylococcus aureus subsp. aureus HO 5096 0412
0.01	10	10	-	1381115	Staphylococcus aureus subsp. aureus Tager 104
0.00	6	6	-	985006	Staphylococcus aureus subsp. aureus LGA251
0.00	4	4	-	685039	Staphylococcus aureus subsp. aureus ED133
0.00	4	4	-	869816	Staphylococcus aureus subsp. aureus JKD6159
0.00	3	3	-	1392476	Staphylococcus aureus subsp. aureus 6850
0.00	2	2	-	1343064	Staphylococcus aureus subsp. aureus ST772-MRSA-V

<3% unclassified

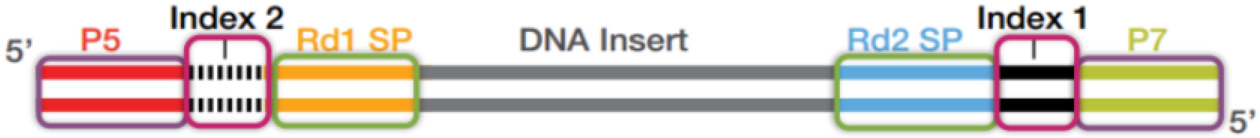
88% S. aureus

Wood DE et al.2014. Kraken: ultrafast metagenomic sequence classification using exact alignments. Genome Biol. doi: 10.1186/gb-2014-15-3-r46.

Wood DE et al. 2019. Improved metagenomic analysis with Kraken 2. Genome Biol. doi: 10.1186/s13059-019-1891-0

TrimGalore

Illumina library:



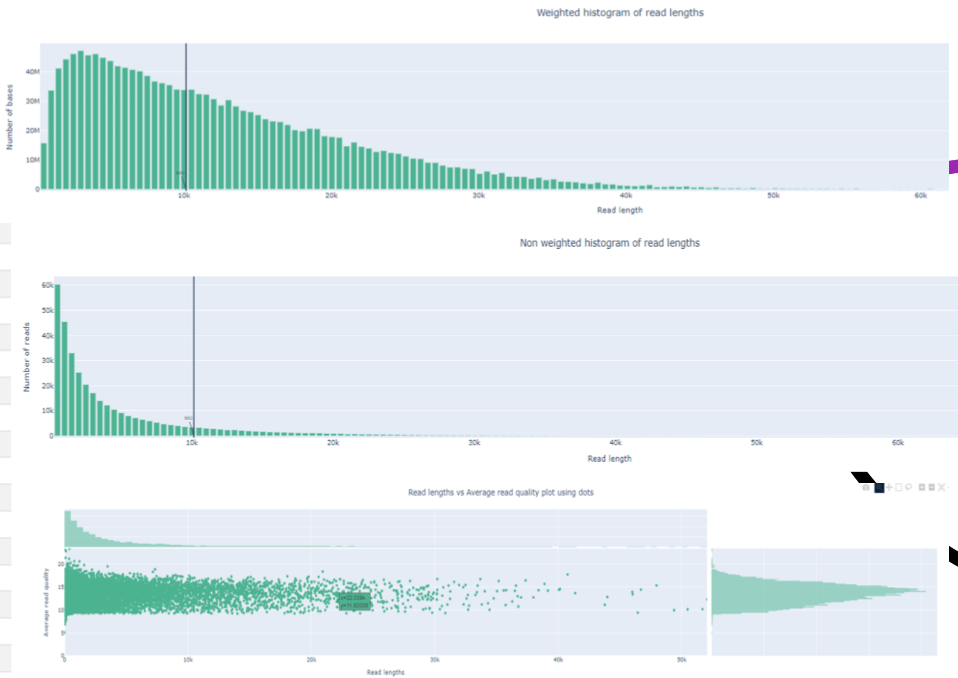
We should trim those bases that were added during library preparation (which are not part of the original sequence) and discard those which have low quality

NanoPlot

NanoPlot reports

Summary statistics

Metrics	
number_of_reads	348283
number_of_bases	1587776501.0
median_read_length	2238.0
mean_read_length	4558.9
read_length_stdev	5913.9
n50	10149.0
mean_qual	12.6
median_qual	13.7
longest_read_(with_Q):1	78693 (10.7)
longest_read_(with_Q):2	70709 (14.6)
longest_read_(with_Q):3	70336 (10.7)
longest_read_(with_Q):4	65788 (11.2)
longest_read_(with_Q):5	64413 (12.1)
highest_Q_read_(with_length):1	27.1 (135)
highest_Q_read_(with_length):2	26.3 (54)
highest_Q_read_(with_length):3	26.3 (56)
highest_Q_read_(with_length):4	25.5 (57)
highest_Q_read_(with_length):5	25.4 (16)
Reads >Q5:	348267 (100.0%) 1587.8Mb
Reads >Q7:	348234 (100.0%) 1587.8Mb
Reads >Q10:	333811 (95.8%) 1521.9Mb
Reads >Q12:	267520 (76.8%) 1228.7Mb
Reads >Q15:	89582 (25.7%) 373.4Mb



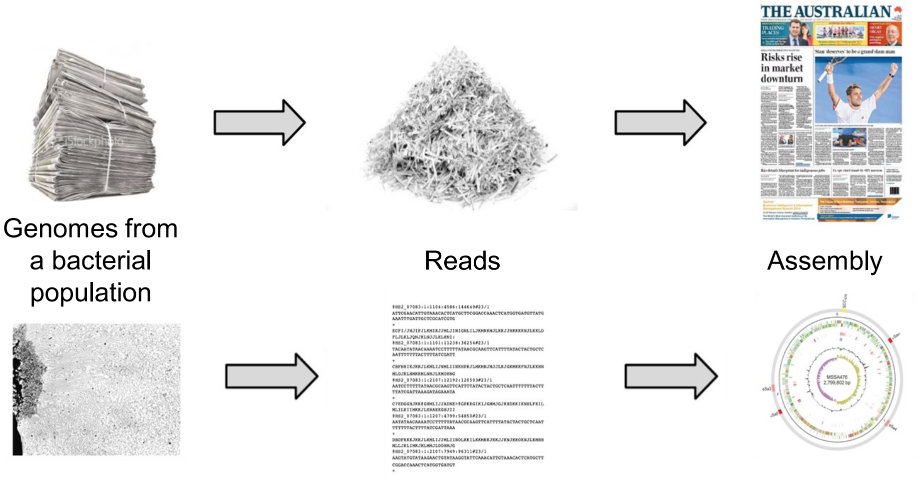
Filtlong

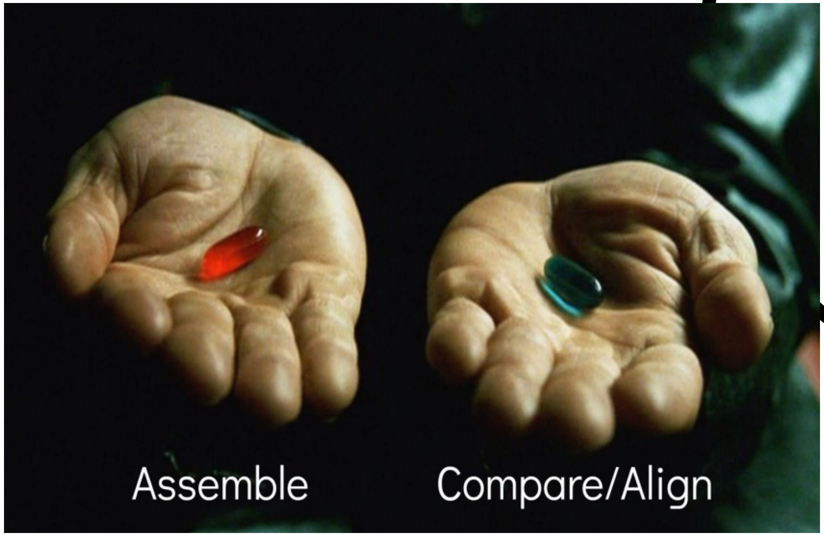
Filtlong

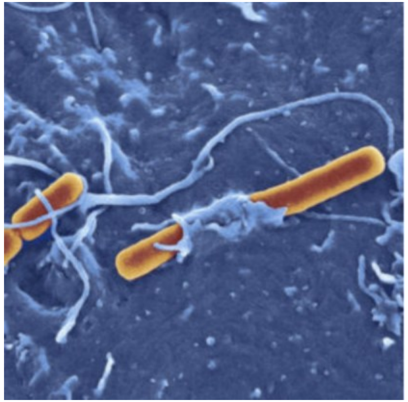
QC filtering can improve the quality of the ONT reads

Tool for filtering long reads by quality. It can take a set of long reads and produce a smaller, better subset.

We have our reads... and now?



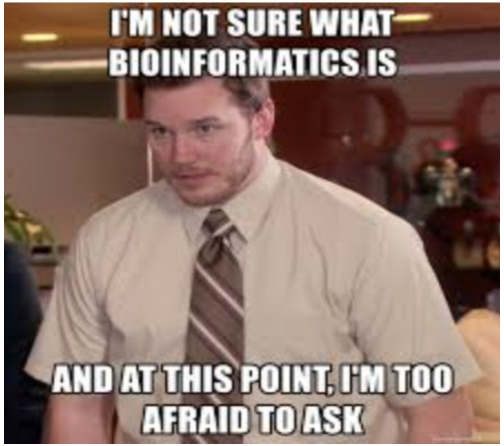




Genomic & Epidemiological
Surveillance of Bacterial Pathogens

22 – 26 June 2026

Thanks for your attention!
Questions?



wellcome connecting science

Wrap up

1. What information did you get about the sequencing runs? (probable platform? Read length?)
2. What do you think about the quality of the sequenced data analysed in this module?
3. Did the trimming solve all the quality issues encountered in all the samples?
4. Would you use these sequences for further analyses? Why?

Basic Statistics

Measure	Value
Filename	CNGB1553_S31_L001_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	528785
Total Bases	132.7 Mbp
Sequences flagged as poor quality	0
Sequence length	251
%GC	49

Basic Statistics

Measure	Value
Filename	CNGB39120_S21_L001_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1718971
Total Bases	224.3 Mbp
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	53

Basic Statistics

Measure	Value
Filename	CNGB1797_S7_L001_R2_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1288460
Total Bases	171.6 Mbp
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	33

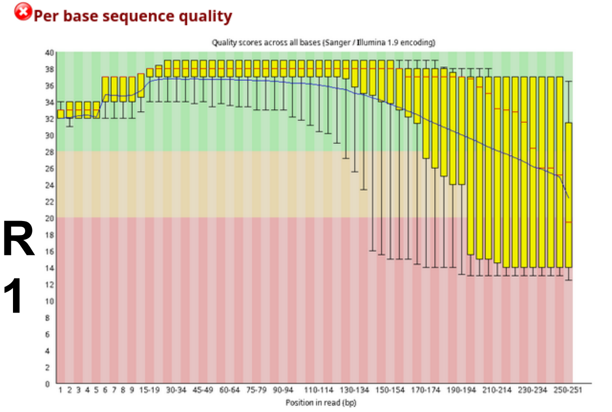
Basic Statistics

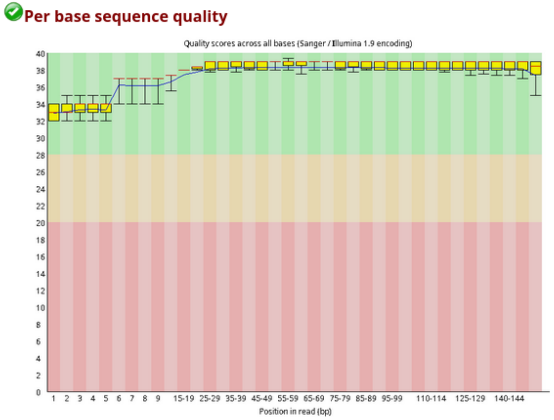
Measure	Value
Filename	CTMA_1441_longds.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	90000
Total Bases	540.3 Mbp
Sequences flagged as poor quality	0
Sequence length	100-107743
%GC	47

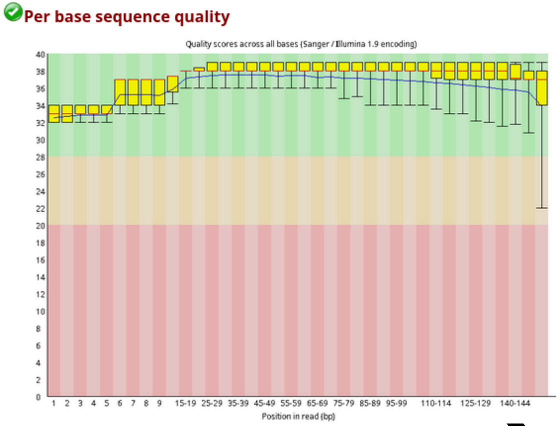
Raw fastq

CNGB1553	CNGB1797	CNGB39120

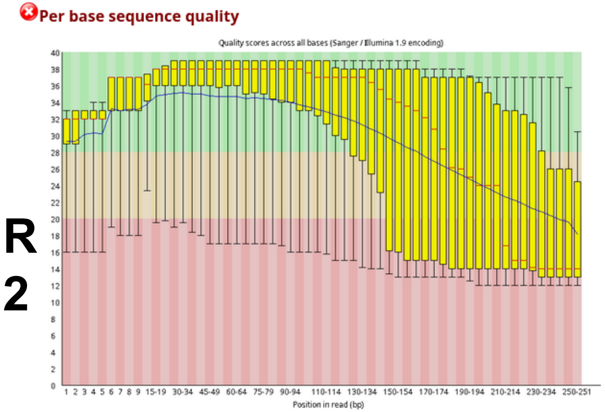
R1

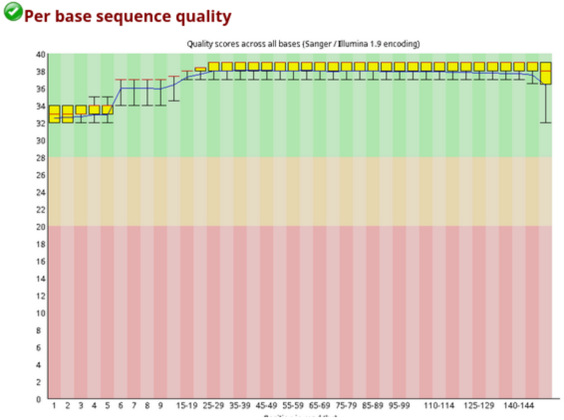


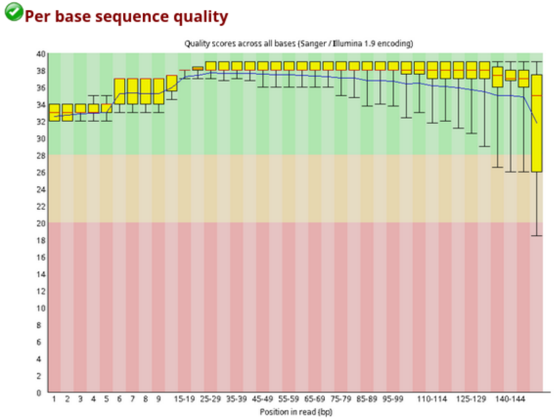




R2







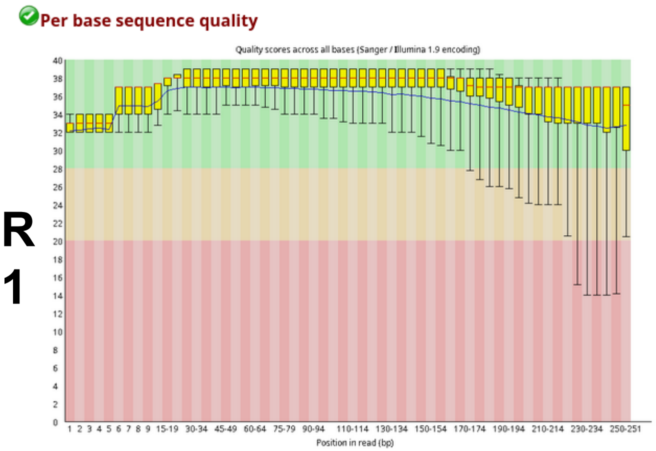
Trimmed fastq

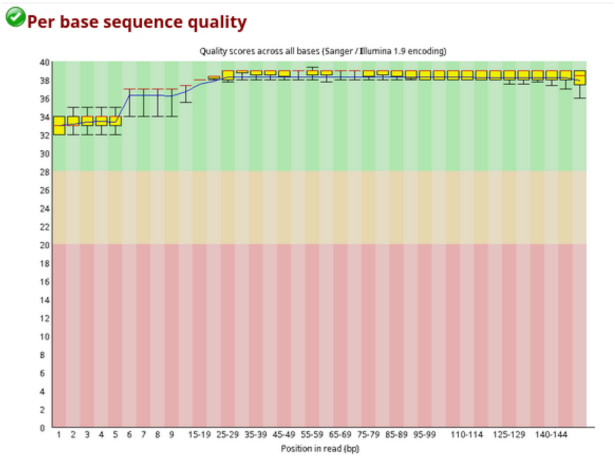
CNGB1553

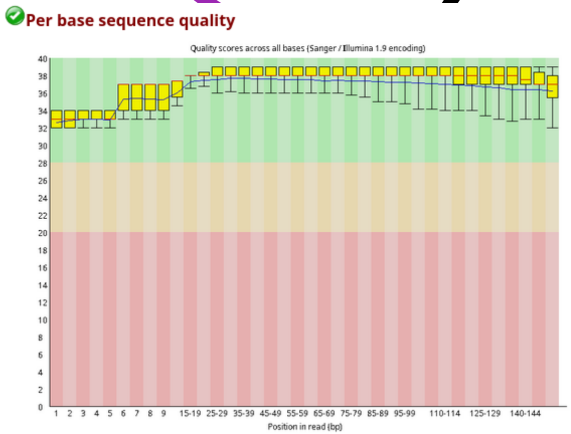
CNGB1797

CNGB39120

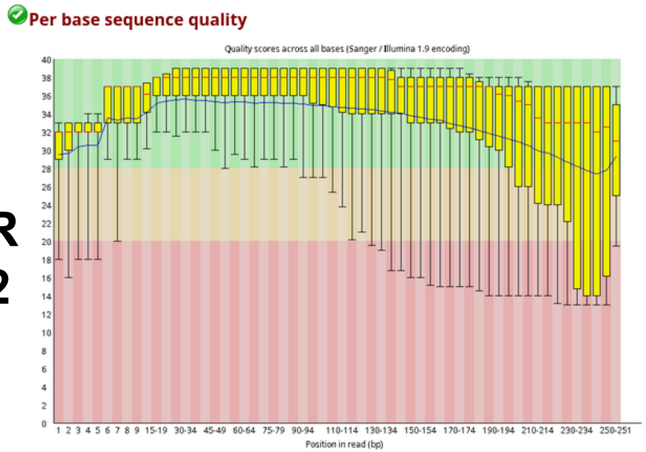
R1

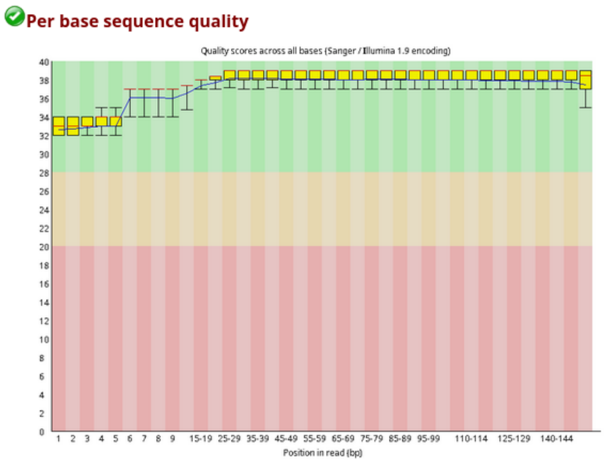


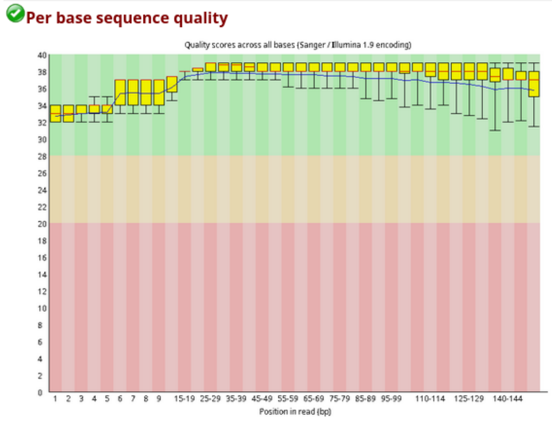




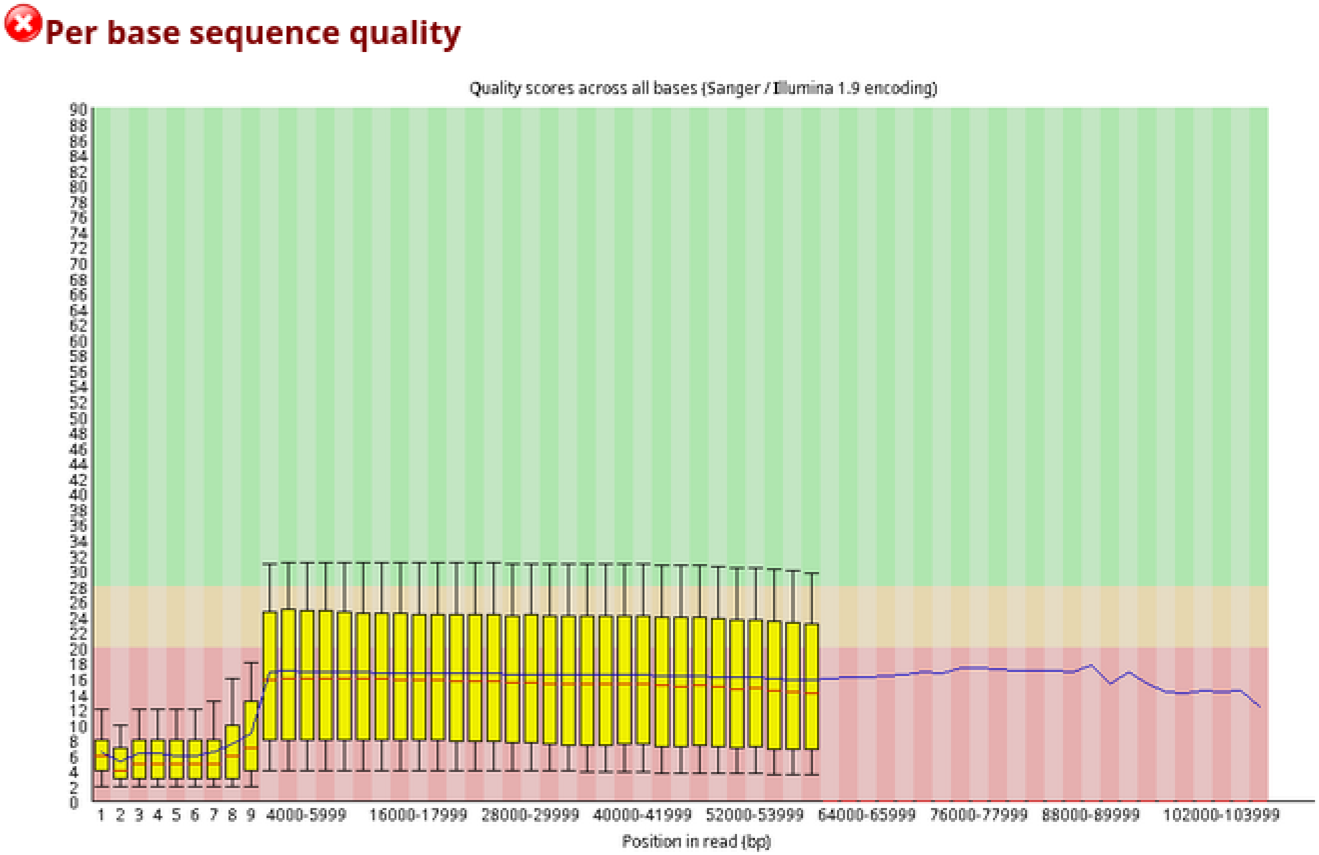
R2





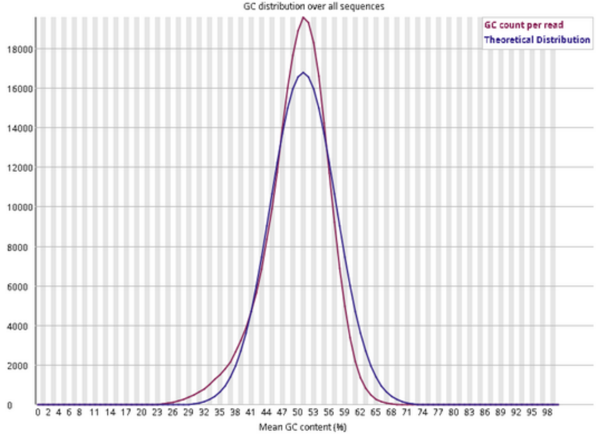


CTMA_1441


Per base sequence quality
Quality scores across all bases (Sanger / Illumina 1.9 encoding)
Position in read (bp)

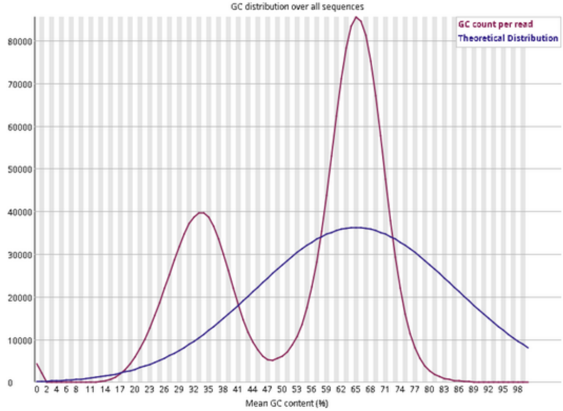
CNGB1553

Per sequence GC content



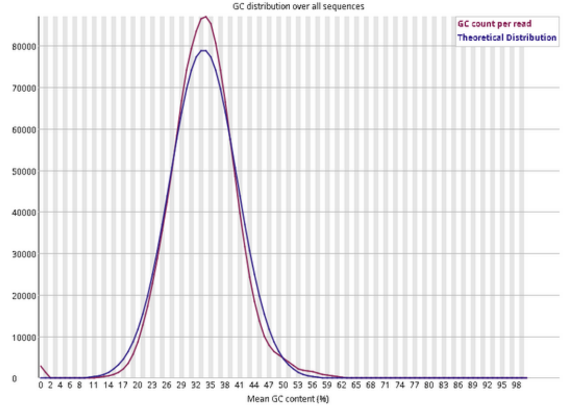
CNGB39120

Per sequence GC content



CNGB1797

Per sequence GC content



CTMA_1441

Per sequence GC content